Dataset Overview

This project utilized multiple datasets from different sources to analyze population distribution and components of population change in Iowa counties. The datasets included county-level population estimates, spatial boundary data, and migration data. Data were collected, cleaned, and integrated to support spatial analysis and visualization in R, GeoDa, ArcGIS, and Tableau.

1. County Population Data

County-level total population data for Iowa were collected using the *tidycensus* package in R from the American Community Survey (ACS) 5-year estimates. The dataset included annual population estimates for all Iowa counties, with particular focus on the five most populous counties identified in the study. Population data from 2013 to 2024 were used to analyze population growth trends over time.

Key Variables Included:

- County Name
- Year
- Total Population
- County Geometry (spatial boundary data)

Purpose in the Project:

- Identify the most populous counties in Iowa
- Analyze population growth trends from 2013–2024
- Create spatial visualizations and choropleth maps
- Conduct Exploratory Spatial Data Analysis (ESDA)

2. Spatial Boundary (Shapefile) Data

Spatial boundary data for Iowa counties were generated using geometry data obtained through *tidycensus* in R. The geometry information was exported as shapefiles using the st_write function and later used in GeoDa, ArcGIS, and Tableau for spatial analysis and mapping.

Key Components Included:

- County Boundaries
- Geographic Coordinates
- Spatial Relationship Information

Purpose in the Project:

- Create county-level maps
- Perform spatial clustering analysis
- Develop interactive dashboards in Tableau

3. Migration Data

County-level migration data were obtained from the IRS migration dataset. The dataset provided estimates of migration flows and net migration for Iowa counties through 2022. These data were cleaned and organized in Excel before analysis.

Key Variables Included:

- County Name
- Year
- In-Migration
- Out-Migration
- Net Migration

Purpose in the Project:

- Analyze migration patterns in selected counties
- Estimate components of population change
- Compare migration trends with overall population growth

4. Derived Natural Increase Dataset

Natural Increase was not directly available in the collected datasets and was therefore calculated during the analysis process using population change and net migration data.

$$\text{Natural Increase} = \text{Population Change} - \text{Net Migration}$$

Key Variables Included:

- County Name
- Year
- Population Change
- Net Migration
- Calculated Natural Increase

Purpose in the Project:

- Estimate the contribution of natural increase to population growth
- Compare the relative influence of migration and natural increase across counties
- Support visualization of components of population change

Data Preparation and Cleaning

All datasets were cleaned and formatted primarily in Excel before analysis. Cleaning procedures included:

- Removing missing or incomplete records
- Standardizing county names and year formats
- Organizing datasets into tabular structures suitable for Tableau visualization
- Creating calculated fields for population change and natural increase

The cleaned datasets were then integrated into Tableau to develop dashboards and story maps that communicate patterns of population distribution and population change in Iowa counties.